Assemblies, constraints & file types

GRAPHICS WITH MR KELLY

National 5 · CAD & technology · Worksheet

Name _____ Class _____ Date _____

By the end of this worksheet I can…

☐ assemble components with mate, align and centre axis

I DO — Worked example

Read how a full-mark answer is built

A wheel is fitted onto an axle and pushed against the side of a bracket, with its outer face level with the bracket's outer face.

1 State the three assembly constraints used. (3 marks)

Full-mark answer each ✓ is one mark

Centre axis, to line the wheel hole up with the axle ✓. Mate, to bring the wheel face against the bracket ✓. Align, to make the outer faces level ✓.

Why marks are not awarded

"Line up, push together and make level."

0 marks: the course specification terms must be used.

WE DO — Guided practice

Use the starters and word bank to help you

An engineer produces orthographic drawings directly from a 3D CAD assembly.

2 Describe two advantages of producing the drawings from the 3D model. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

accurate · always · dimensions · drawings · changed · straight · update · automatically

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Two faces touching, facing each other, is...
☐ Align ☐ Mate ☐ Centre axis ☐ Shell

b) Two faces level with each other, facing the same way, is...
☐ Align ☐ Mate ☐ Centre axis ☐ Chamfer

c) Which constraint lines up a pin with a hole?
☐ Mate ☐ Align ☐ Centre axis ☐ Fillet

4 Key terms: write the letter of the matching description next to each term.

1. **Mate**	____	**A** Two faces made flush, facing the same way.
2. **Align**	____	**B** Lines up the axes of two round features.
3. **Centre axis**	____	**C** Two faces brought together facing each other.

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?